CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 16: write the python program to implement feed forward neural network

Aim:

To write a Python program to implement a Feed Forward Neural Network and predict outputs using a given dataset.

Algorithm:

1. Start the program.
2. Import the required libraries, such as NumPy and Scikit-learn.
3. Define the input dataset and target output.
4. Create a feed-forward neural network with an input layer, a hidden layer, and an output layer.
5. Train the neural network using the given dataset.
6. Predict the outputs using the trained model.
7. Compare the predicted outputs with the actual outputs.
8. Display the predicted results and accuracy.
9. Stop the program.

py16.py - C:/Users/navit/py16.py (3.12.10)

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```
import math

def sigmoid(x):
    return 1.0 / (1.0 + math.exp(-x))

inputs = [[0, 0], [0, 1], [1, 0], [1, 1]]

print("Feed Forward Neural Network")
print("----------------------------")

for data in inputs:
    h1 = sigmoid(data[0] * 0.5 + data[1] * 0.5 - 0.2)
    h2 = sigmoid(data[0] * 0.4 + data[1] * 0.4 - 0.1)

    output = sigmoid(h1 * 0.6 + h2 * 0.6 - 0.3)

    print("Input:", data, "Output:", round(output, 4))
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/User
Feed Forward Neural Network
----------------------------
Input: [0, 0] Output: 0.5634
Input: [0, 1] Output: 0.5961
Input: [1, 0] Output: 0.5961
Input: [1, 1] Output: 0.626
>>>
```

Result:

The Feed Forward Neural Network was implemented successfully using Python. The model was trained on the given dataset, and the predicted outputs and classification accuracy were displayed.